CHAPTER

4

REAL FUNCTIONS OF REAL VARIABLES

4.1 Introduction

Let $D \in \mathbb{R}$ i.e D is a subset of the real numbers. Often we need to associate with $x \in D$ a new real number which we denote at the moment by $f(x)$. For example the absolute value.

Definition 4.1. *Let $D \in \mathbb{R}$. A function $f : D \longmapsto R$ is a rule which assigns to every $x \in D$ exactly one real value $f(x)$. For this we write $x \longmapsto f(x)$ and say that x is mapped onto $f(x)$, or $f(x)$ is the value of f at x.*

We call D the domain of the function f, sometimes we write $D(f)$ or D_f instead of D. So $D_f = \{x \in D : f(x) \text{ exists}\}$.

The set of all functions is denoted by $\mathcal{F}(D, \mathbb{R})$.

Definition 4.2. *The set $\{y = f(x), x \in D\}$ is called the range of f and is denoted by $H(f)$. Variable x is called argument or independent variable and variable y is called dependent.*

Definition 4.3. *Let $f, g \in \mathcal{F}(D, \mathbb{R})$ and $\lambda \in \mathbb{R}$. We define the following important operations:*

1. $(f+g)(x) = f(x) + g(x)$ $(\forall x \in D)$
2. $(f.g)(x) = f(x).g(x)$ $(\forall x \in D)$
3. $(\lambda g)(x) = \lambda g(x)$ $(\forall \lambda \in \mathbb{R}, \forall x \in D)$
4. *If* $\forall x \in D : f(x) \neq 0$, $\left(\frac{1}{f}\right)(x) = \frac{1}{f(x)}$.

Definition 4.4. *Let f and g be real functions with domains $D(f)$ and $D(g)$. Let $H(f) \subset D(g)$. Then under the composition of function f and g we understand function h defined by $\forall x \in D(h) : h(x) = g(f(x))$, with D(h) = D(f).*

Notation: $h = gof$.

Definition 4.5. *Two functions f and g are equal $(f = g)$, if*
(i) $D(f) = D(g)$
(ii) $\forall x \in D(f) : f(x) = g(x)$

Remark 4.1. *In general $gof \neq fog$.*

Definition 4.6. *Graph of function f is a set of ordered pairs of real numbers $(x, f(x))$, where $x \in D(f)$. We write*

$$graphf = \{(x, f(x))/x \in D(f)\}$$

Even and odd functions

Definition 4.7. *Let $D \subset \mathbb{R}$ such that $(\forall x \in D) \Longrightarrow (-x \in D)$*
We say that function $f : D \longrightarrow \mathbb{R}$ is even if and only if $(\forall x \in D), (f(-x) = f(x))$.
We say that function $f : D \longrightarrow \mathbb{R}$ is odd if and only if $(\forall x \in D), (f(-x) = -f(x))$.

Periodic functions

Definition 4.8. *A function $f : D \longrightarrow \mathbb{R}$ is called periodic if $\exists \quad T > 0$ such that*
– $\forall x \in D \Longrightarrow x \pm T \in D, f(x+T) = f(x)$
– $\forall x \in D, f(x+T) = f(x)$. *Number T is called a period of f. The smallest positive period is called primitive.*

Remark 4.2. *Let $T > 0$ be a period of f, then*

$$\forall x \in \mathbb{R}, \forall n \in \mathbb{Z} : f(x + nP) = f(x).$$

Theorem 4.1. *(i) If f is periodic with period P and function g such that $H(f) \subset D(g)$ then a composition $h(x) = g(f(x))$ is periodic with the same period p. (ii) If f is periodic with period p and $a \in \mathbb{R}, a \neq 0$; then function $g(x) = f(ax)$ is periodic with period $\frac{P}{a}$.*

Proof. Just do it. $\square$

Bounded Functions

Definition 4.9. *Let $f : D \to \mathbb{R}$ and let $f(D)$ the set of all the values of f.*

We say that function f is bounded above on its domain D if $f(D)$ is bounded above i.e. $\exists M \in \mathbb{R}, \forall x \in D : f(x) \leqslant M$.
We say that function f is bounded below on its domain D if $f(D)$ is bounded below i.e. $\exists m \in \mathbb{R}, \forall x \in D : f(x) \geqslant m$.
We say that function f is bounded on its domain D if $f(D)$ is bounded i.e. $\exists A \in \mathbb{R}, \forall x \in D : |f(x)| \leqslant A$.

Remark 4.3. *If f is bounded on D, then $f(D)$ admits a supremum M and an infimum m. We denote $M = \sup\limits_{x\in D} f(x)$, and $m = \inf\limits_{x\in D} f(x)$. We have*

$$\sup_{x\in D} f(x) = M < +\infty \Longleftrightarrow \begin{cases} \forall x \in D : f(x) \leqslant M \\ \forall \varepsilon > 0, \exists x_0 \in D : f(x_0) > M - \varepsilon \end{cases}$$

$$\inf_{x\in D} f(x) = m > -\infty \Longleftrightarrow \begin{cases} \forall x \in D : f(x) \geqslant m \\ \forall \varepsilon > 0, \exists x_0 \in D : f(x_0) < m + \varepsilon \end{cases}$$

Monotone functions

Definition 4.10. *Consider $f : D \subset \mathbb{R} \to \mathbb{R}$, and set $M \subset D$.*

1) f is nondecreasing on $M \Leftrightarrow \forall x_1, x_2 \in M,\ x_1 \leqslant x_2 \Longrightarrow f(x_1) \leqslant f(x_2)$
2) $f$is nonincreasing on $M \Leftrightarrow \forall x_1, x_2 \in M,\ x_1 \leqslant x_2 \Longrightarrow f(x_1) \geqslant f(x_2)$
3) f is increasing on $M \Leftrightarrow \forall x_1, x_2 \in M,\ x_1 < x_2 \Longrightarrow f(x_1) < f(x_2)$
4) f is decreasing on $M \Leftrightarrow \forall x_1, x_2 \in M,\ x_1 < x_2 \Longrightarrow f(x_1) > f(x_2)$

Definition 4.11. *If f satisfies any of condition $(1-4)$ we call it monotone. If f has property (3) or (4), we call it strictly monotone.*

Corollary 4.1. *We can check that in case where f is increasing or decreasing then we have $x_1 \neq x_2 \Longrightarrow f(x_1) \neq f(x_2)$, so f is injective.*
A sum of two increasing (decreasing) functions is an increasing (decreasing) function.

Inverse functions

Definition 4.12. *Let $f : D(f) \longleftrightarrow \mathbb{R}$ be an injective function with range $H(f)$. Inverse function of f (denoted f^{-1}) is defined by the relation $y = f(x) \Longleftrightarrow x = f^{-1}(y)$. Obviously the domain $D(f^{-1}) = H(f)$ and range $H(f^{-1}) = D(f)$.*

Remark 4.4. *(i) Graph of f^{-1} is symmetric to the graph of f with respect to a line $y = x$.*
(ii) $\forall x \in D(f) : f^{-1}(f(x)) = x$.
(iii)$\forall y \in D(f^{-1}) = H(f) : f(f^{-1}(y)) = y$.
(iv) $(f^{-1})^{-1} = f$.

Lemma 4.2. *Let A, B be two subset of $\mathbb{R}$. Let $f : A \longrightarrow B$ be a bejective and strictly monotonic function. The f^{-1} is strictly monotonic function, the same monotonic as f.*

Proof. WLOG, we can suppose that f is strictly increasing. let y_1, y_2 be two elements of B such that $y_1 < y_2$, then we prove that $f^{-1}(y_1) < f^{-1}(y_2)$. Since f is bejective then there exist x_1, x_2 such that $f(x_1) = y_1 \quad \text{and} \quad f(x_2) = y_2$. Let us proceed by contra-positive, we suppose that $x_1 \geq x_2$ then $y_1 = f(x_1) \geq f(x_2) = y_2$ which is a contradiction with $y_1 < y_2$. □

Lemma 4.3. *Let I be an interval of $\mathbb{R}$ and $f : I \longrightarrow \mathbb{R}$ be a strictly monotonic function such that $f(I)$ is an interval, then f is necessary continuous on I.*

Proof. WLOG, we can suppose that I is not a trivial interval $I = \emptyset, \quad I = \{a\}$ and we suppose that f is strictly increasing. Let $a \in I$, if $a > inf(I)$, we prove that $\lim\limits_{x \to a^-} f(x) = f(a)$ and if $a < sup(I)$, we prove that $\lim\limits_{x \to a^+} f(x) = f(a)$. Indeed let $a \in I$, if $a > inf(I)$, by monotonic limit theorem, there exists a real l such that $\lim\limits_{x \to a^-} f(x) = l$, and $l \leq f(a)$. Actually, $l = sup\{f(x), x \in I, x < a\}$, we aim to prove that $l = f(a)$. By contrapositive, suppose that $l < f(a)$, then there exists $l < m < f(a)$, there exists $\alpha \in I$ such that $\alpha < a$. By the hypothesis that f is increasing, we get $f(\alpha) \leq l < m < f(a)$. $f(I)$ is an interval, so there exists $c \in I$ such that $f(c) = m$.

- If $c \geq a$, and f is supposed to be increasing, hence $f(c) \geq f(a) > m$,
- If $c < a$, monotonic limit gives $f(c) < l < m$, which is a contradiction, thus $f(a) = l$.

Similarly, with the right limit. □

Lemma 4.4. *Let I be an interval of $\mathbb{R}$ and $f : I \longrightarrow \mathbb{R}$ be a continuous and injective function. Then f is strictly monotonic.*

Proof. By contrapositive principal, we suppose that f isn't strictly monotone, thus

- $\exists x, y \in I : x < y$ and $f(x) \geq f(y)$
- $\exists x', y' \in I : x' < y'$ and $f(x') \leq f(y')$

The segments $[x, x'], [y, y']$ are defined by $[x, x'] = \{tx + (1-t)x', t \in [0,]\}$ and $[y, y'] = \{ty + (1-t)y', t \in [0, 1]\}$. Then let us define the following functions:

$$\alpha : [0, 1] \longrightarrow \mathbb{R} : t \to tx + (1-t)x',$$

$$\beta : [0, 1] \longrightarrow \mathbb{R} : t \to ty + (1-t)y'.$$

$\alpha(t)$ and $\beta(t)$ belong to the interval I. Now we consider the function defined by

$$\phi : [0, 1] \longrightarrow \mathbb{R} : t \to f(\alpha(t)) - f(\beta(t)).$$

(1) α, β, f are continuous so ϕ,
(2) $\phi(0) = f(x) - f(y) \geq 0$ and $\phi(1) = f(x') - f(y') \leq 0$. The mean theorem value implies that there exists $t_0 \in]0, 1[$ such that $\phi(t_0) = 0$, which means that $f(\alpha(t_0) = f(\beta(t_0)$. Contradiction (f is injective). □

The inverse function theorem for strictly monotonic function

Theorem 4.5. *Let I be an interval of $\mathbb{R}$ and $f : I \longrightarrow \mathbb{R}$ be a function. Set $J = f(I)$. Then two of the following properties implie the third one.*

1- J is an interval and $f : I \longrightarrow J$ is a bejection function.

2- f is strictly monotonic on I.

3- f is continuous on I.

more; if 1, 2 and 3 are satisfied, then the inverse function $f^{-1} : J \longrightarrow I$ is continuous, strictly monotonic, the same as f.

Proof.
- If 1 and 2 are satisfied then f is continuous (Lemma (4.26)).
- If 1 and 3 are satisfied then f is strictly monotonic (Lemma (4.27)).
- If 2 and 3 are satisfied then J is an interval (MTV theorem). f is strictly monotonic, thus f is injective and by the way bejective.

If 1, 2 and 3 are satisfied the by (Lemma (4.25)), $f^{-1} : J \longrightarrow I$ is strictly monotonic, the same as f. $f^{-1} : J \longrightarrow I$ realize a bejection from J on I, so f^{-1} satisfies 1 and 2, hence f^{-1} is continuous. □

Inverse image

Let $f : A \longrightarrow B$ be a function, and let $U \subset B$ be a subset. The inverse image (or, preimage) of U is the set $f^{-1}(U) \subset A$ consisting of all elements $a \in A$ such that $f(a) \in U$.

The inverse image commutes with all set operations: For any collection $\{U_i\}_{i\in I}$ of subsets of B, we have the following identities for

(1) Unions:

$$f^{-1}\left(\bigcup_{i\in I} U_i\right) = \bigcup_{i\in I} f^{-1}\left(U_i\right)$$

(2) Intersections:

$$f^{-1}\left(\bigcap_{i\in I} U_i\right) = \bigcap_{i\in I} f^{-1}\left(U_i\right)$$

and for any subsets U and V of B, we have identities for

(3) Complements:

$$\left(f^{-1}(U)\right)^{\complement} = f^{-1}\left(U^{\complement}\right)$$

(4) Set differences:

$$f^{-1}(U \backslash V) = f^{-1}(U) \backslash f^{-1}(V)$$

(5) Symmetric differences:

$$f^{-1}(U \triangle V) = f^{-1}(U) \triangle f^{-1}(V)$$

In addition, for $X \subset A$ and $Y \subset B$, the inverse image satisfies the miscellaneous identities

(6) $\left(f|_X\right)^{-1}(Y) = X \cap f^{-1}(Y)$

(7) $f\left(f^{-1}(Y)\right) = Y \cap f(A)$

(8) $X \subset f^{-1}(f(X))$, with equality if f is injective. Let $f : A \longrightarrow B$ be a function, and let $U \subset A$ be a subset. The direct image (or, simply, image) of U is the set $f(U) \subset B$ consisting of all elements of B which equal $f(u)$ for some $u \in U$.

Direct images satisfy the following properties:

(1) Unions: For any collection $\{U_i\}_{i\in I}$ of subsets of A,

$$f\left(\bigcup_{i\in I} U_i\right) = \bigcup_{i\in I} f\left(U_i\right).$$

(2) Intersections: For any collection $\{U_i\}_{i\in I}$ of subsets of A,

$$f\left(\bigcap_{i\in I} U_i\right) \subset \bigcap_{i\in I} f\left(U_i\right).$$

(3) Set difference: For any $U, V \subset A$,

$$f(V\backslash U) \supset f(V)\backslash f(U).$$

In particular, the complement of U satisfies $f\left(U^{\complement}\right) \supset f(A)\backslash f(U)$.

(4) Subsets: If $U \subset V \subset A$, then $f(U) \subset f(V) \subset B$.

(5) Inverse image of a direct image: For any $U \subset A$,

$$f^{-1}(f(U)) \supset U$$

with equality if f is injective.

(6) Direct image of an inverse image: For any $V \subset B$,

$$f\left(f^{-1}(V)\right) \subset V$$

with equality if f is surjective.

Local maximum, local minimum Local maximum and minimum are the points of the functions, which give the maximum and minimum range. The local maxima and local minima can be computed by finding the derivative of the function. The first derivative test and the second derivative test are the two important methods of finding the local maximum and local minimum.

Definition 4.13. *Let $f : D \subset \mathbb{R} \to \mathbb{R}$ and $x_0 \in E$.*
– x_0 is said to be local maximum of f if there exists $\alpha > 0$ such that f is nondecreasing on $]x_0 - \alpha, x_0[$ and nonincreasing on $]x_0, x_0 + \alpha[$.
– x_0 is said to be local minimum of f if there exists $\alpha > 0$ such that f is nonincreasing on $]x_0 - \alpha, x_0[$ and nondecreasing on $]x_0, x_0 + \alpha[$.

Order relation on $\mathcal{F}(D, \mathbb{R})$.

Definition 4.14. *Let $f, g : D \subset \mathbb{R} \to \mathbb{R}$.*
1) f is said to be positive $f \geqslant 0$, if : $\forall x \in D : f(x) \geqslant 0$ (resp. negative, $f \leqslant 0$ if $\forall x \in D : f(x) \leqslant 0$.
2) f is said to be greater then $g, (f \geqslant g)$, if: $\forall x \in D : f(x) \geqslant g(x)$.
3) f is said to be smaller then $g, (f \leqslant g)$, if: $\forall x \in D : f(x) \leqslant g(x)$.

Remark 4.5. *We can easily check that this order relation isn't total.*

4.2 Limit of a Function

The basic idea underlying the concept of the limit of a function f at a point x_0 is to study the behaviour of f at points close to, but not equal to, x_0. We illustrate this with the following simple examples. Suppose that the velocity $v(m/s)$ of a falling object is given as a function $v = v(t)$ of time t. If the object hits the ground in $t = 2$, then $v(2) = 0$. Thus to find the velocity at the time of impact, we investigate the behaviour of $v(t)$ as t approaches 2, but is not equal to 2. Neglecting air resistance, the function $v(t)$ is given as follows :

$$v(t) = \begin{cases} 32t, 0 \leqslant t < 2, \\ 0, t \geqslant 2. \end{cases}$$

Our intuition should convince us that $v(t)$ approaches $64m/s$ as t approaches 2 , and that this is the velocity upon impact. As another example, consider the function $f(x) = xsin\left(\frac{1}{x}\right)$, $x \neq 0$. Here the function f is not defined at $x = 0$. Thus to investigate the behaviour of f at 0 we need to consider the values $f(x)$ for x close to, but not equal to 0 . Since

$$|f(x)| = \left|xsin\left(\frac{1}{x}\right)\right| \leqslant |x|$$

for all $x \neq 0$, our intuition again should tell us that $f(x)$ approaches 0 as x approaches 0. We now make this idea of (x) approaching a value L as x approaches a point x_0 precise. In order that the definition be meaningful, we must require that the point p be a limit point of the domain of the function f .

Definition 4.15. *Let $x_0 \in \mathbb{R}$ an accumulation point of a subset $D \neq \phi \subset \mathbb{R}$ and $f : D \mapsto \mathbb{R}$ a function defined on a neighbourhood of x_0 except may be at x_0. The function f has a limit at x_0 if there exists $l \in \mathbb{R}$ such that*

$$\forall \varepsilon > 0, \exists \delta\left(x_0, \varepsilon\right) > 0, \forall x : 0 < |x - x_0| < \delta \Rightarrow |f\left(x\right) - l| < \varepsilon$$

Customary we write: $\lim\limits_{x \to x_0} f\left(x\right) = l$ or $f\left(x\right) \to l,\ x \to x_0$
Shortly

$$\lim_{x \to x_0} f\left(x\right) = l \Leftrightarrow \forall \varepsilon > 0, \exists \delta\left(x_0, \varepsilon\right) > 0, \forall x : 0 < |x - x_0| < \delta \Rightarrow |f\left(x\right) - l| < \varepsilon$$

Uniqueness of the limit

Theorem 4.6. *If f admits a limit then it is unique.*

Proof. Just do it. □

One-Sided Limits

It is possible for a function to fail to have a limit at a point and yet appear to have limits on one side. If we ignore what is happening on the right for a function, perhaps it will have a " left-hand limit." This is easy to achieve

Definition 4.16. *We say that a function f has right-hand limit (resp. left-hand limit) at x_0 if $\forall\varepsilon > 0, \exists\delta > 0, \forall x : x_0 < x < x_0 + \delta$ (resp. $x_0 - \delta < x < x_0$) $\Rightarrow |f(x) - l| < \varepsilon$ we note*

$$\lim_{x \to x_0 + 0} f(x) = f(x_0 + 0), \lim_{x \to x_0 - 0} f(x) = f(x_0 - 0).$$

We have

$$\lim_{x \to x_0} f(x) = l \Longleftrightarrow \lim_{x \to x_0 + 0} f(x) = \lim_{x \to x_0 - 0} f(x) = l$$

Limit at infinity

Definition 4.17. *Let $x_0 = +\infty$ be an accumulation point of a given subset D. Then*

$$\lim_{x \to +\infty} f(x) = l \Leftrightarrow \forall\varepsilon > 0, \exists A > 0, \forall x > A \Rightarrow |f(x) - l| < \varepsilon$$

Let $x_0 = -\infty$ be an accumulation point of a given subset D. Then

$$\lim_{x \to -\infty} f(x) = l \Leftrightarrow \forall\varepsilon > 0, \exists A > 0, \forall x < -A \Rightarrow |f(x) - l| < \varepsilon$$

Infinite limits

Definition 4.18. *Let $x_0 \in \mathbb{R}$ be an accumulation point of a given subset D and $f : D \to \mathbb{R}$. Then*

$$\lim_{x \to x_0} f(x) = +\infty \Leftrightarrow \forall A > 0, \exists\delta > 0, \forall x : 0 < |x - x_0| < \delta \Rightarrow f(x) > A$$

$$\lim_{x \to x_0} f(x) = -\infty \Leftrightarrow \forall A > 0, \exists\delta > 0, \forall x : 0 < |x - x_0| < \delta \Rightarrow f(x) < -A$$

Main limit's theorems

Our first theorem allows us to reduce the question of the existence of the limit of a function to one concerning the existence of limits of sequences. As we will see, this result will be very useful in subsequent proofs, and also in showing that a given function does not have a limit at a point x_0.

Theorem 4.7. *Let $f : D \subset \mathbb{R} \to \mathbb{R}$ be a given function and x_0 be an accumulation point of a given subset D. Then*

1) $\lim\limits_{x\to x_0} f(x) = l$
if and only if
2) For all sequence $(x_n)_{n\in\mathbb{N}}$, $x_n \in D/\{x_0\}$ *and* $\lim\limits_{n\to+\infty} x_n = x_0$ *we obtain* $\lim\limits_{n\to+\infty} f(x_n) = l$, *(l finite or not.)*

Proof. Since x_0 is a limit point of D, there exists a sequence $x_n \in D$ with $x_n \neq x_0$ for all $n \in \mathbb{N}$ and $\lim\limits_{n\to+\infty} x_n = x_0$.

Suppose $\lim\limits_{x\to x_0} f(x) = l$. Let x_n be any sequence in D with $x_n \neq x_0$ for all $n \in \mathbb{N}$ and $\lim\limits_{n\to+\infty} x_n = x_0$. Let $\varepsilon > 0$ be given. Since $\lim\limits_{x\to x_0} f(x) = l$, there exists a $\delta > 0$ such that

$$|f(x) - l| < \varepsilon \quad \text{for all} \quad x \in D, 0 < |x - x_0| < \delta. \tag{4.1}$$

Since $\lim\limits_{n\to+\infty} x_n = x_0$, for the above δ, there exists a positive integer n_0 such that $|x_n - x_0| < \delta$, for all $n \geqslant n_0$. Thus if $n \geqslant n_0$, by (4.1), $|f(x_n) - l| < \varepsilon$. Therefore $\lim\limits_{n\to+\infty} f(x_n) = l$.
Conversely, suppose that $\mathop{lim}\limits_{n\to+\infty} f(x_n) = l$ for every sequence $x_n \in D$ with $x_n \neq x_0$ for all $n \in \mathbb{N}$ and $\lim\limits_{n\to+\infty} x_n = x_0$. Suppose $\lim\limits_{x\to x_0} f(x) \neq l$. Then there exists an $\varepsilon > 0$ such that for every $\delta > 0$, there exists an $x \in D$ with $0 < |x - x_0| < \delta$ and $|f(x) - l| \geqslant \varepsilon$. For each $n \in \mathbb{N}$, take $\delta = \frac{1}{n}$. Then for each n, there exists $x_n \in D$ such that

$$|x_n - x_0| < \frac{1}{n} \quad \textit{and} \quad |f(x_n) - l| \geqslant \varepsilon.$$

Thus $x_n \longrightarrow x_0$, but $f(x_n)$ does not converge to l. This contradiction proves the result. □

Some limit laws We now state some limit laws for functions.

Theorem 4.8. *Let* $f, g : D \subset \mathbb{R} \to \mathbb{R}$ *be real functions, and* x_0 *an accumulation (cluster) point of* D. *Suppose that* $\lim\limits_{x\to x_0} f(x) = l_1$, $\lim\limits_{x\to x_0} g(x) = l_2$, $l_1, l_2 \in \mathbb{R}$.

Then

1) $\lim\limits_{x\to x_0} (f(x) \pm g(x)) = l_1 + l_2$
2) $\lim\limits_{x\to x_0} (\lambda f(x)) = \lambda l_1$ $(\lambda \in \mathbb{R})$
3) $\lim\limits_{x\to x_0} (f(x) g(x)) = l_1 l_2$
4) $\lim\limits_{x\to x_0} |f(x)| = |l|$
5) $\lim\limits_{x\to x_0} \dfrac{f(x)}{g(x)} = \dfrac{l_1}{l_2}$, *if* $l_2 \neq 0$.

Proof. The proofs are left as an exercises. (To prove the results, use the sequential criterion for limits and the limits laws for sequences). □

Theorem 4.9. *Let $f, g : D \subset \mathbb{R} \to \mathbb{R}$ be functions such that $\forall x \in D : f(x) \leqslant g(x)$. Suppose that $\lim_{x \to x_0} f(x) = l_1$, and $\lim_{x \to x_0} g(x) = l_2$. then $l_1 \leqslant l_2$.*

Proof. Set $F(x) = g(x) - f(x)$ and $L = l_2 - l_1$. It is sufficient to prove that $L \geqslant 0$. We prove the contrapositive. Suppose then that $L < 0$. Let $\varepsilon > 0$ be such that $L + \varepsilon < 0$. Then since $\lim_{x \to x_0} (g - f)(x) = L$, there exists $\delta > 0$ such that if $0 < |x - x_0| < \delta$ then $(g - f)(x) = L + \varepsilon < 0$. Hence, $(g - f)(x) < 0$ for some $x \in D$. We can give another proof using the sequential criterion for limits. $\square$

Corollary 4.2. *$f : D \subset \mathbb{R} \to \mathbb{R}$ be a function and let x_0 be an accumulation point of D. Suppose that $M_1 \leqslant f(x) \leqslant M_2$ for all $x \in D$ and suppose that $\lim_{x \to x_0} f(x) = L$. Then $M_1 \leqslant L \leqslant M_2$.*

Corollary 4.3. *Let $f : D \subset \mathbb{R} \to \mathbb{R}$ be a function and x_0 be an accumulation point of D. If $\lim_{x \to x_0} f(x) = l \in \mathbb{R}$, then $\exists \delta > 0$ such that for $0 < |x - x_0| < \delta$ the functionf is bounded.*

Theorem 4.10. *Let $f, g : D \subset \mathbb{R} \to \mathbb{R}$ be a functions and x_0 be an accumulation point of D. If g is bounded on D and $\lim_{x \to x_0} f(x) = 0$, then $\lim_{x \to x_0} f(x) g(x) = 0$.*

Proof. Use squeeze technique. $\square$

Theorem 4.11. *Suppose that the limit $\lim_{x \to x_0} f(x) = l$, exists. Then $\lim_{x \to x_0} |f(x)| = |l|$.*

Since maxima and minima can be expressed in terms of absolute values, there is a corollary that is sometimes useful.

Corollary 4.4. *Suppose that the limits $\lim_{x \to x_0} f(x) = L$, and $\lim_{x \to x_0} g(x) = M$, exist and that x_0 is a point of accumulation of $D_f \cap D_g$. Then $\lim_{x \to x_0} max\{f(x), g(x)\} = max\{L, M\}$, and $\lim_{x \to x_0} min\{f(x), g(x)\} = min\{L, M\}$.*

Proof. The first follows from the identity $max\{f(x), g(x)\} = \frac{f(x)+g(x)}{2} + \frac{|f(x)-g(x)|}{2}$ and the theorem on limits of sums and the theorem on limits of absolute values. In the same way the second assertion follows from $min\{f(x), g(x)\} = \frac{f(x)+g(x)}{2} - \frac{|f(x)-g(x)|}{2}$. $\square$

4.3 Continuous functions

Let f be a function defined on an interval $[a, b]$. We shall now consider the behaviour of f at points of $[a, b]$.

Continuity at a Point

Definition 4.19. *A function f is said to be continuous at a point $x_0, a < x_0 < b$, if*

$$\lim_{x \to x_0} f(x) = f(x_0)$$

In other words, the function is continuous at x_0, if for each $\varepsilon > 0, \exists \delta > 0$, such that

$$|f(x) - f(x_0)| < \varepsilon, \text{ when } |x - x_0| < \delta$$

Definition 4.20. *A function f is said to be continuous from the left at x_0, if*

$$\lim_{x \to x_0 - 0} f(x) = f(x_0)$$

Also f is continuous from the right at x_0, if

$$\lim_{x \to x_0 + 0} f(x) = f(x_0)$$

Clearly a function is continuous at x_0 if and only if it is continuous from the left as well as from the right.

Definition 4.21. *A function f defined on a closed interval $[a, b]$ is said to be continuous at the end point a if it is continuous from the right at a,*

i.e.,

$$\lim_{x \to a+0} f(x) = f(a)$$

Also the function is continuous at the end point b of $[a, b]$ if

$$\lim_{x \to b-0} f(x) = f(b)$$

Thus a function f is continuous at a point x_0 if

(i) $\lim_{x \to x_0} f(x)$ exists, and

(ii) limit equals the value of the function at $x = x_0$.

Continuity in an Interval

A function f is said to be continuous in an interval $[a, b]$ if it is continuous at every point of the interval.

Discontinuous Functions

A function is said to be discontinuous at a point x_0 of its domain if it is not continuous there. The point x_0 is then called a point of discontinuity of the function.

4.4 Types of discontinuities

(i) A function f is said to have a removable discontinuity at $x = x_0$ if $\lim_{x\to x_0} f(x)$ exists but is not equal to the value $f(x_0)$ (which may or may not exist) of the function. Such a discontinuity can be removed by assigning a suitable value to the function at $x = x_0$.

(ii) f is said to have a discontinuity of the first kind at $x = x_0$ if $\lim_{x\to x_0-0} f(x)$ and $\lim_{x\to x_0+0} f(x)$ both exist but are not equal.

(iii) f is said to have a discontinuity of the first kind from the left at $x = x_0$ if $\lim_{x\to x_0-0} f(x)$ exists but is not equal to $f(x_0)$

Discontinuity of the first kind from the right is similarly defined.

(iv) f is said to have a discontinuity of the second kind at $x = x_0$ if neither $\lim_{x\to x_0-0} f(x)$ nor $\lim_{x\to x_0+0} f(x)$ exists.

(v) f is said to have a discontinuity of the second kind from the left at $x = x_0$ if $\lim_{x\to x_0-0} f(x)$ does not exist.

Discontinuity of the second kind from the right may be defined similarly.

Theorems on Continuity

Theorem 4.12. *A function f defined on an interval I is continuous at a point $x_0 \in I$ if for every sequence $\{c_n\}$ in I converging to x_0, we have*

$$\lim_{n\to\infty} f(c_n) = f(x_0)$$

Proof. First let us suppose that the function f is continuous at a point $x_0 \in I$, and $\{c_n\}$ is a sequence in I such that $\lim_{n\to\infty} c_n = x_0$.

Since f is continuous at x_0, therefore, for any given $\varepsilon > 0, \exists$ a $\delta > 0$, such that

$$|f(x) - f(x_0)| < \varepsilon, \text{ when } 0 < |x - x_0| < \delta \tag{4.2}$$

Again, since $\lim_{n\to\infty} c_n = x_0$, therefore, $\exists$ a positive integer m, such that

$$|c_n - x_0| < \delta, \forall n \geq m \tag{4.3}$$

From (4.2), putting $x = c_n$, we have

$$\begin{array}{rl} & |f(c_n) - f(x_0)| < \varepsilon, \text{ when } |c_n - x_0| < \delta \\ \Rightarrow & |f(c_n) - f(x_0)| < \varepsilon, \quad \forall n \geq m \text{ [using 2]} \\ \Rightarrow & \text{the sequence } \{f(c_n)\} \text{ converges to } f(x_0) \end{array}$$

or

$$\lim_{n\to\infty} f(c_n) = f(x_0)$$

Let us now suppose that f is not continuous at x_0, we shall now show that though there exists a sequence (c_n) in I converging to x_0 yet the sequence $(f(c_n))$ does not converge to $f(x_0)$.

Since f is not continuous at x_0, therefore, there exists an $\varepsilon > 0$ such that for every $\delta > 0$, there exists an $x \in I$, such that

$$|f(x) - f(x_0)| \geq \varepsilon, \text{ when } |x - x_0| < \delta$$
$$\lim c_n = x_0.$$

Also by taking $\delta = 1/n$, we find that for each positive integer n, there is a $c_n \in I$, such that

$$|f(c_n) - f(x_0)| \geq \varepsilon, \text{ when } |c_n - x_0| < \frac{1}{n}$$

Thus, the sequence $(f(c_n))$ does not converge to $f(x_0)$, while the sequence (c_n) converges to x_0. □

Remark 4.6. *If* $\lim c_n = x_0 \Rightarrow \lim f(c_n) \neq f(x_0)$, *then* f *is not continuous at* x_0.

Theorem 4.13. *If* f, g *be two functions continuous at a point* x_0, *then the functions* $f+g, f-g, fg$ *are also continuous at* x_0 *and if* $g(x_0) \neq 0$, *then* f/g *is also continuous at* x_0.

The proof is left as an exercise (use the sequential criterion for limits and the laws for sequences).

Example 4.1. *Examine the following function for continuity at the origin:*

$$f(x) = \begin{cases} \frac{xe^{1/x}}{1+e^{1/x}} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Now

$$\lim_{x\to 0-} f(x) = \lim_{x\to 0-} \frac{xe^{1/x}}{1+e^{1/x}} = 0$$

and

$$\lim_{x\to 0+} f(x) = \lim_{x\to 0+} \frac{x}{e^{-1/x}+1} = 0$$

Thus,

$$\lim_{x\to 0-} f(x) = \lim_{x\to 0+} f(x) = \lim_{x\to 0} f(x) = 0$$

Also

$$\lim_{x\to 0} f(x) = f(0)$$

Thus, the function is continuous at the origin.

Example 4.2. *Show that the function defined as:*

$$f(x) = \begin{cases} \frac{\sin 2x}{x}, & when\ x \neq 0 \\ 1, & when\ x = 0 \end{cases}$$

has removable discontinuity at the origin.

Solution. *Now*

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{\sin 2x}{2x} \cdot 2 = 2$$

so that

$$\lim_{x\to 0} f(x) \neq f(0)$$

Hence, the limit exists, but is not equal to the value of the function at the origin. Thus the function has a removable discontinuity at the origin.

Note. The discontinuity can be removed by re-defining the function at the origin such as $f(0) = 2$.

Example 4.3. *Show that the function defined by*

$$f(x) = \begin{cases} x\sin 1/x, & when\ x \neq 0 \\ 0, & when\ x = 0 \end{cases}$$

is continuous at $x = 0$.

Solution. *Now*

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} \left(x \sin\frac{1}{x}\right) = 0$$

so that

$$\lim_{x \to 0} f(x) = f(0)$$

Hence, f is continuous at $x = 0$.

Example 4.4. *A function f is defined on $[0, 1]$ by*

$$f(x) = \begin{cases} -x^2 & \textit{if } x \leq 0 \\ 5x - 4 & \textit{if } 0 < x \leq 1 \\ 4x^2 - 3x & \textit{if } 1 < x < 2 \\ 3x + 4 & \textit{if } x \geq 2 \end{cases}$$

Examine f for continuity at $x = 0, 1, 2$. Also discuss the kind of discontinuity, if any.
Solution. Now

$$\lim_{x \to 0-} f(x) = \lim_{x \to 0-} \left(-x^2\right) = 0$$
$$\lim_{x \to 0+} f(x) = \lim_{x \to 0+} (5x - 4) = -4$$

so that

$$\lim_{x \to 0-} f(x) = f(0) \neq \lim_{x \to 0+} f(x)$$

Thus the function has a discontinuity of the first kind from the right at the origin.
Again

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} (5x - 4) = 1$$
$$\lim_{x \to 1+} f(x) = \lim_{x \to 1+} \left(4x^2 - 3x\right) = 1$$

so that

$$\lim_{x \to 1-} f(x) = \lim_{x \to 1+} f(x) = 1 = f(1)$$
$$\lim_{x \to 1} f(x) = f(1)$$

Thus the function is continuous at $x = 1$.
Again

$$\lim_{x \to 2-} f(x) = \lim_{x \to 2-} \left(4x^2 - 3x\right) = 10$$
$$\lim_{x \to 2+} f(x) = \lim_{x \to 2+} (3x + 4) = 10$$

Also $f(2) = 10$

$\Rightarrow$

$$\lim_{x \to 2} f(x) = f(2)$$

Thus, the function is continuous at $x = 2$.

Example 4.5. *Is the function f, where $f(x) = \frac{x-|x|}{x}$ continuous?*

Solution. *For $x < 0, f(x) = \frac{x+x}{x} = 2$, continuous.*

For $x > 0, f(x) = \frac{x-x}{x} = 0$, continuous. The function is not defined at $x = 0$.

Thus, $f(x)$ is continuous for all x except zero.

Example 4.6. *Discuss the kind of discontinuity, if any of the function defined as follows:*

$$f(x) = \begin{cases} \frac{x-|x|}{x}, & \text{when } x \neq 0 \\ 2, & \text{when } x = 0 \end{cases}$$

Solution. *The function is continuous at all points except possibly the origin.*

Let us test at $x = 0$.

Now

$$\lim_{x \to 0-} f(x) = \lim_{x \to 0-} \frac{x+x}{x} = 2$$
$$\lim_{x \to 0+} f(x) = \lim_{x \to 0+} \frac{x-x}{x} = 0$$

and

$$f(0) = 2$$

Thus the function has discontinuity of the first kind from the right at $x = 0$.

Example 4.7. *If $[x]$ denotes the largest integer $\leq x$, then discuss the continuity at $x = 3$ for the function*

$$f(x) = x - [x], \quad \forall x \geq 0$$

Solution. *Now*

$$\lim_{x \to 3-} f(x) = \lim_{x \to 3-} \{x - [x]\} = 3 - 2 = 1$$
$$\lim_{x \to 3+} f(x) = \lim_{x \to 3+} \{x - [x]\} = 3 - 3 = 0$$

and

$$f(3) = 0$$

Thus the function has a discontinuity of the first kind from the left at $x = 3$.

Note. The function is continuous at all points except the integer value $1, 2, 3, \ldots$

Example 4.8. *Prove that the Dirichlet's function f defined on $\mathbf{R}$ by*

$$f(x) = \begin{cases} 1, & \text{when } x \text{ is irrational} \\ -1, & \text{when } x \text{ is rational} \end{cases}$$

is discontinuous at every point.

Solution. First, let a be any rational number so that $f(a) = -1$.

Since in any interval there lie an infinite number of rational and irrational numbers, therefore, for each positive integer n, we can choose an irrational number a_n, such that $|a_n - a| < \frac{1}{n}$.

Thus, the sequence $\{a_n\}$ converges to a. But $f(a_n) = 1$ for all n, and $f(a) = -1$, so that

$$\lim_{n \to \infty} f(a_n) \neq f(a)$$

Thus, the function is discontinuous at any rational number a.

Hence the function is discontinuous at all rational points.

Next, let b be any irrational number. For each positive integer n we can choose a rational number b_n, such that $|b_n - b| < \frac{1}{n}$. Thus, the sequence (b_n) converges to b.

But $f(b_n) = -1$ for all n and $f(b) = 1$.

Therefore $\lim_{n \to \infty} f(b_n) \neq f(b)$.

Hence, the function is discontinuous at all irrational points.

Example 4.9. *Show that the function $f(x)$ defined on $\mathbb{R}$ by*

$$f(x) = \begin{cases} x, & \text{when } x \text{ is irrational} \\ -x, & \text{when } x \text{ is rational} \end{cases}$$

is continuous only at $x = 0$.

Solution. First, let $a \neq 0$ be any rational number, so that $f(a) = -a$. Since in every interval there lie an infinite number of rational and irrational numbers, therefore, for each positive integer n, we can choose an irrational number a_n such that

Thus the sequence (a_n) converges to a.

$$|a_n - a| < \frac{1}{n}$$

But

$$\lim_{n\to\infty} f(a_n) = \lim_{n\to\infty} a_n = a$$

Thus

$$\lim_{n\to\infty} f(a_n) \neq f(a), a \neq 0$$

so that, the function is discontinuous at any rational number, other than zero.

In a similar way the function may be shown to be discontinuous at every irrational point.

It may be seen from above, that the function is continuous at $x = 0$ (i.e., $a = 0$). However, it can be shown to be continuous at $x = 0$ as follows:

Let $\varepsilon > 0$ be given and let $\delta = \varepsilon$ (or any $\delta < \varepsilon$), then

$$|x| < \delta \Rightarrow |f(x) - f(0)| = |-x| = |x| < \varepsilon, \text{ when } x \text{ is rational and}$$
$$|x| < \delta \Rightarrow |f(x) - f(0)| = |x| < \varepsilon, \text{ when } x \text{ is irrational.}$$

Thus

$$|x| < \delta \Rightarrow |f(x) - f(0)| < \varepsilon$$

or

$$\lim_{x\to 0} f(x) = f(0)$$

Hence, the function is continuous at $x = 0$.

4.4.1 Continuous functions on closed intervals

We shall now study some properties of functions which are continuous on closed intervals. In fact, we shall show that a function which is continuous on a closed interval, is bounded, attains its bounds and assumes every value between the bounds.

Theorem 4.14. *If a function is continuous in a closed interval, then it is bounded therein.*

Proof. Let f be a function defined and continuous in a closed interval I.

We shall show that if the function f is not bounded, then it fails to be continuous at some point of the closed interval I

Let, if possible, f be not bounded above, so that for each positive integer $n \exists$ a point $x_n \in I$ such that $f(x_n) > n$

Now $\{x_n\}$, being a sequence in the closed interval I, is bounded and has at least one limit point, say ξ.

A closed interval is a closed set and so $\xi \in I$.

Further, since ξ is a limit point of the sequence (x_n), therefore, there exists (Bolzano-Weirestrass theorem) a subsequence (x_{n_k}) of (x_n) such that $x_{n_k} \to \xi$ as $k \to \infty$.

Also since $f(x_{n_k}) > n_k$, for all k, therefore the sequence $(f(x_{n_k}))$ diverges to ∞.

Thus, there exists a point ξ of I such that a sequence (x_{n_k}) in I converges to ξ, but

$$\lim_{k \to \infty} f(x_{n_k}) \neq f(\xi)$$

Thus, f is not continuous at ξ, which is a contradiction and hence the function is bounded above. By considering a function $-f$, it can be shown in a similar way that the function f is also bounded below. Hence, the function is bounded. □

Theorem 4.15. *If a function f is continuous on a closed interval $[a, b]$, then it attains its bounds at least once in $[a, b]$.*

Proof. If f is a constant function, then evidently, it attains its bounds at every point of the interval.

Let f be a function which is not a constant.

Since f is continuous on the closed interval $[a, b]$, therefore, it is bounded. Let m and M be the infimum and supremum of f. It is to be shown that $\exists$ points, α, β of $[a, b]$ such that

$$f(\alpha) = m, \quad f(\beta) = M$$

Let us consider the case of the supremum.

Suppose f does not attain the supremum M so that the function does not take the value M for any point $x \in [a, b]$, i.e.,

$$f(x) \neq M, \text{ for any } x \in [a, b]$$

Now consider the function

$$g(x) = \frac{1}{M - f(x)}, \quad \forall x \in [a, b]$$

which is positive for all values of x in $[a, b]$.

Evidently the function g is continuous and so bounded in $[a, b]$.

Let $k (> 0)$ be its supremum

$$\begin{aligned} & \tfrac{1}{M-f(x)} \leq k, \forall x \in [a,b] \\ \Rightarrow \quad & f(x) \leq M - \tfrac{1}{k}, \forall x \in [a,b] \end{aligned}$$

which contradicts the hypothesis that M is the supremum of f in $[a,b]$. Hence, our supposition that f does not attain the value M leads to a contradiction and therefore f attains its supremum for at least one value in $[a,b]$.

It may similarly be shown that the function also attains its infimum m.

Hence, the function attains its bounds at least once in $[a,b]$.

$\square$

Note It may be observed from the two preceding theorems, that the function f, continuous on the closed interval $[a,b]$, has the least and the greatest values m and M, i.e., the range set of f is bounded with m and M as its smallest and greatest elements. Thus the range set of f is a subset of $[m,M]$. We shall, in fact, show later that the range set of f is $[m,M]$ itself and that f takes up every value between m and M.

4.4.2 Examples

1. The function $f(x) = \frac{1}{1+|x|}$, for real x, is continuous and bounded and attains its supremum for $x = 0$ but does not attain the infimum.

2. The function $f(x) = -\frac{1}{1+|x|}$, for all $x \in \mathbb{R}$, is continuous and bounded, attains its infimum but not the supremum.

3. The function $f(x) = x$, for all $x \in]0,1[$ is continuous and bounded but attains neither the infimum nor the supremum.

Theorem 4.16. *If a function f is continuous at an interior point c of an interval $[a,b]$ and $f(c) \neq 0$, then $\exists$ a $\delta > 0$ such that $f(x)$ has the same sign as $f(c)$, for every $x \in]c-\delta, c+\delta[$.*

Proof. Since the function f is continuous at an interior point c of $[a,b]$, therefore for any $\varepsilon > 0, \exists$ a $\delta > 0$, such that

$$|f(x) - f(c)| < \varepsilon, \quad \forall x \in]c-\delta, c+\delta[$$

or

$$f(c) - \varepsilon < f(x) < f(c) + \varepsilon, \quad \forall x \in]c-\delta, c+\delta[$$

When $f(c) > 0$, taking ε to be less than $f(c)$, we find that

$$f(x) > 0, \forall x \in]c-\delta, c+\delta[$$

When $f(c) < 0$, taking ε to be less than $-f(c)$, we find that

$$f(x) < 0, \forall x \in]c-\delta, c+\delta[$$

Hence the theorem.

□

Corollary 4.5. *If f is continuous at the end point b of $[a,b]$ and $f(b) \neq 0$, then there exists an interval $]b-\delta, b]$, such that $f(x)$ has the sign of $f(b)$ for all x in $]b-\delta, b]$*

A similar result holds for continuity at a.

Note When c is an interior point of the interval, the theorem may be restated as:

Theorem 4.17. *If a function f is continuous at an interior point c of an interval $[a,b]$ and $f(c) \neq 0$, then there exists a neighbourhood N of c wherein $f(x)$ has the same sign as $f(c)$, for all $x \in N$.*

Theorem 4.18. *(intermediate value theorem) If a function f is continuous on a closed interval $[a,b]$ and $f(a)$ and $f(b)$ are of opposite signs, then there exists at least one point $\alpha \in]a,b[$ such that $f(\alpha) = 0$.*

Proof. Let us suppose that $f(a) > 0$ and $f(b) < 0$.

Let S consist of those points of $[a,b]$ for which $f(x)$ is positive, i.e.,

$$S = \{x : a \leq x \leq b \wedge f(x) > 0\}$$

Now

$$f(a) > 0 \Rightarrow a \in S \Rightarrow S \neq \phi$$

Also S is bounded by a and b.

By the order completeness property, S has the supremum, say α, where $a \leq \alpha \leq b$. We shall now show that

(i) $\alpha \neq a, \alpha \neq b$, and

(ii) $f(\alpha) = 0$

(i) First we show that $\alpha \neq a$

Since $f(a) > 0$, therefore $\exists$ a $\delta > 0$ such that

$$f(x) > 0, \quad \forall x \in [a, a+\delta[$$

$\Rightarrow \quad [a, a+\delta[\subseteq S$

$\Rightarrow$ the supremum α of S is greater than or equal to $a+\delta$

$\Rightarrow \alpha \neq a$

Now we shall show that $\alpha \neq b$.

Since $f(b) < 0$, therefore $\exists$ a $\delta > 0$ such that

$$f(x) < 0, \quad \forall x \in]b-\delta, b]$$

$\Rightarrow b-\delta$ is an upper bound of S

$\Rightarrow \alpha \leq b-\delta \Rightarrow \alpha \neq b$

(ii) We shall now show that $f(\alpha) \not> 0$ and $f(\alpha) \not< 0$.

If $f(\alpha) > 0$, then $\exists$ a $\delta > 0$ such that

$$f(x) > 0, \quad \forall x \in]\alpha-\delta, \alpha+\delta[$$
$$\Rightarrow \quad]\alpha-\delta, \alpha+\delta[\subseteq S$$

Let us choose a positive $\delta_2 < \delta$ such that $\alpha+\delta_2 \in]\,\alpha-\delta, \alpha+\delta\,[\Rightarrow$ a member $\alpha+\delta_2$ of S is greater than the supremum α of S, which is a contradiction.

Therefore

$$f(\alpha) \not> 0$$

Let now $f(\alpha) < 0$, so that $\exists$ a $\delta_1 > 0$ such that

$$f(x) < 0, \quad \forall x \in]\alpha-\delta_1, \alpha+\delta_1[\tag{4.4}$$

Again, since α is the supremum of S, therefore, there exists a member β of S, where $\alpha-\delta_1 < \beta \leq \alpha$ such that

$$f(\beta) > 0$$

But from (4.4), $f(\beta) < 0$, which is a contradiction.

Therefore $\quad f(\alpha) \not< 0$

Thus it follows that $f(\alpha) = 0$.

□

Theorem 4.19. *If a function f is continuous on $[a, b]$ and $f(a) \neq f(b)$, then it assumes every value between $f(a)$ and $f(b)$*

Proof. Let A be any number between $f(a)$ and $f(b)$. We shall show that there exists a number $c \in]a, b[$ such that $f(c) = A$. Consider a function ϕ defined on $[a, b]$ such that

$$\phi(x) = f(x) - A$$

Clearly $\phi(x)$ is continuous on $[a, b]$.

Also

$$\phi(a) = f(a) - A, \text{ and } \phi(b) = f(b) - A$$

so that $\phi(a)$ and $\phi(b)$ are of opposite signs.

Thus the function ϕ is continuous on $[a, b]$ and $\phi(a)$ and $\phi(b)$ are of opposite signs; therefore, by the previous theorem, $\exists c \in]a, b[$ such that

$$\phi(c) = 0 \Longrightarrow f(c) - A \quad = 0 \Rightarrow f(c) = A,$$

□

Corollary 4.6. *A function f, which is continuous on a closed interval $[a, b]$, assumes every value between its bounds.*

Proof. Since the function f is continuous on the closed interval $[a, b]$, therefore, it is bounded and attains its bounds on $[a, b]$, i.e., $\exists$ two numbers α, β in $[a, b]$ such that

$$f(\alpha) = M \text{ and } f(\beta) = m,$$

where M and m are, respectively, the supremum and the infimum of f.

Since f is continuous on $[a, b]$, therefore, it is continuous on $[\beta, \alpha]$ or $[\alpha, \beta]$ as the case may be, and consequently assumes every value between $f(\alpha)$ and $f(\beta)$.

Thus the function assumes every value between its bounds.

We may sum up in other words:

The range of a continuous function whose domain is a closed interval is as well a closed interval. Or, in still better words:

The image of a closed interval under a continuous function (mapping) is a closed interval.

□

4.4.3 Uniform continuity

Let f be a function defined on an interval I. Then by definition, the function is continuous at any point $x_0 \in I$ if for any $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - f(x_0)| < \varepsilon, \text{ when } |x - x_0| < \delta.$$

For continuity at any other point $d \in I$, for the same ε, a $\delta_1 > 0$ would exist (not necessarily equal to δ). There is in fact a δ corresponding to each point of I. The number δ in general depends on the selection of ε and the point x_0. However, if a δ could be found which depends only on ε and not on the selection of the point x_0, such a δ would work for the whole interval I on which f is continuous. In such a case, f is said to be uniformly continuous on I. Thus, the notion of uniform continuity is global in character in as much as we talk of uniform continuity only on an interval.

The notion of continuity is, however, local in character in as much as we can talk of continuity at a point.

It may seem to a beginner that the infimum of the set consisting of δ 's corresponding to different points of I would work for the whole of I. But the infimum may be zero. In general, therefore, a δ which may work for the entire interval may not exist, so that every continuous function may not be uniformly continuous.

Definition 4.22. *A function f defined on an interval I is said to be uniformly continuous on I if to each $\varepsilon > 0$ there exists a $\delta > 0$ such that*

$$|f(x_2) - f(x_1)| < \varepsilon, \text{ for arbitrary points } x_1, x_2 \text{ of } I$$
$$\text{for which } |x_1 - x_2| < \delta$$

Theorem 4.20. *A function which is uniformly continuous on an interval is continuous on that interval.*

Let a function f be uniformly continuous on an interval I, so that for a given $\varepsilon > 0$, there corresponds a $\delta > 0$ such that

$|f(x_1) - f(x_2)| < \varepsilon$, where x_1, x_2 are any two points of I for which

$$|x_1 - x_2| < \delta$$

Let $x \in I$, then on taking $x_1 = x$, we find that for $\varepsilon > 0, \exists \delta > 0$ such that

$$|f(x) - f(x_2)| < \varepsilon, \text{ when } |x - x_2| < \delta.$$

Hence the function is continuous at every point $x_2 \in I$, i.e., the function f is continuous on I.

Theorem 4.21. *A function which is continuous on a closed interval is also uniformly continuous on that interval.*

Proof. Let a function f be continuous on a closed interval I. Let, if possible, f be not uniformly continuous on I. Then there exists an $\varepsilon > 0$ such that for any $\delta > 0$, there are numbers $x, y \in I$ for which

$$|f(x) - f(y)| \nless \varepsilon, \text{ when } |x - y| < \delta$$

In particular for each positive integer n, we can find real numbers x_n, y_n in I such that

$$|f(x_n) - f(y_n)| \nless \varepsilon, \text{ when } |x_n - y_n| < 1/n$$

Now (x_n) and (y_n) being sequences in the closed interval I, they are bounded and so each has at least one limit point, say ξ and η respectively.

As a closed interval is a closed set,

therefore

$$\xi \in I, \eta \in I$$

Further since ξ is a limit point of (x_n), there exists a convergent subsequence (x_{n_k}) of (x_n), such that $x_{n_k} \to \xi$.

Similarly, there exists a convergent subsequence (y_{n_k}) of (y_n) such that $y_{n_k} \to \eta$.

Again from above, we find that

$$|f(x_{n_k}) - f(y_{n_k})| \nless \varepsilon, \text{ when } |x_{n_k} - y_{n_k}| < 1/n_k \leq 1/k$$

The second inequality shows that

$$\lim_{k\to\infty} x_{n_k} = \lim_{k\to\infty} y_{n_k}$$
$$\xi = \eta$$

From the first inequality we find that in case the sequences $(f(x_{n_k}))$ and $(f(y_{n_k}))$ converge, the limits to which they converge are different.

We thus have two sequences (x_{n_k}) and (y_{n_k}) both of which converge to ξ but $(f(x_{n_k}))$ and $(f(y_{n_k}))$ do not converge to the same limit.

So f is not continuous at ξ, for, otherwise, the two sequences $(f(x_{n_k}))$ and $(f(y_{n_k}))$ would converge to the same point $f(\xi)$.

Thus we arrive at a contradiction and so the hypothesis that f is not uniformly continuous on I is false.

Hence f is uniformly continuous on I. □

Theorem 4.22. *If $f, g : (a, b) \longrightarrow \mathbb{R}$ are both uniformly continuous, then $f + g$ and $f - g$ (as functions from (a, b) to $\mathbb{R}$) are also uniformly continuous. $fg : (a, b) \longrightarrow \mathbb{R}$ is uniformly continuous.*

Theorem 4.23. *Let f and g be uniformly continuous on $\mathbb{R}$. Then their composition*

$f \circ g$ is also uniformly continuous on $\mathbb{R}$.

Definition 4.23. *A function $f : \mathbb{R} \longrightarrow \mathbb{R}$ is defined to be Lipschitz if there exists a constant $K > 0$ such that for all $a, b \in \mathbb{R}$, we have $|f(a) - f(b)| \leq K|a - b|$*

Theorem 4.24. *A Lipschitz function is uniformly continuous. A periodic and continuous function is uniformly continuous.*

Example 4.10. *Show that the function $f(x) = 1/x$ is not uniformly continuous on $]0, 1]$.*

Solution. *Clearly the function is continuous on $]0, 1]$.*

It will be uniformly continuous on the given interval if for a given $\varepsilon > 0, \exists$ a $\delta > 0$, independent of the choice of points x and c in $]0, 1]$, such that

$$\left|\frac{1}{x} - \frac{1}{c}\right| < \varepsilon, \text{ when } |x - c| < \delta$$

or

$$\left|\frac{c - x}{cx}\right| < \varepsilon, \text{ when } c - \delta < x < c + \delta \tag{4.5}$$

If we take $c = \delta$, then the interval $]c - \delta, c + \delta[$ becomes $]0, 2\delta[$. Also condition (4.5) must hold for any x in this interval.

But

$$\frac{\delta - x}{\delta x} \to \infty \text{ as } x \to 0,$$

i.e., if we choose x sufficiently close to zero, then condition (4.5)) is violated.

Hence $1/x$ is not uniformly continuous on $]0, 1]$.

Example 4.11. *Show that the function $f(x) = x^2$ is uniformly continuous on $[-1, 1]$.*

Solution. *Let x_1, x_2 be any two points of $[-1, 1]$, then*

$$\begin{aligned} &|f(x_1) - f(x_2)| = \left|x_1^2 - x_2^2\right| = |x_1 - x_2| \cdot |x_1 + x_2| < \varepsilon \\ &\text{when } |x_1 - x_2| < \frac{1}{2}\varepsilon = \delta \end{aligned}$$

(where δ is independent of the choice of x_1, x_2).

Thus for any $\varepsilon > 0, \exists$ a $\delta = \frac{1}{2}\varepsilon$ such that for any choice of x_1, x_2 in $[-1, 1]$, we have

$$|f(x_1) - f(x_2)| < \varepsilon, \text{ when } |x_1 - x_2| < \frac{1}{2}\varepsilon = \delta$$

Thus the function f is uniformly continuous on $[-1, 1]$.

Inverse functions

Lemma 4.25. *Let A, B be two subset of $\mathbb{R}$. Let $f : A \longrightarrow B$ be a bejective and strictly monotonic function. The f^{-1} is strictly monotonic function, the same monotonic as f.*

Proof. WLOG, we can suppose that f is strictly increasing. let y_1, y_2 be two elements of B such that $y_1 < y_2$, then we prove that $f^{-1}(y_1) < f^{-1}(y_2)$. Since f is bejective then there exist x_1, x_2 such that $f(x_1) = y_1$and $\quad f(x_2) = y_2$. Let us proceed by contrapositive, we suppose that $x_1 \geq x_2$ then $y_1 = f(x_1) \geq f(x_2)y_2$ which is a contradiction with $y_1 < y_2$. □

Lemma 4.26. *let I be an interval of $\mathbb{R}$ and $f : I \Longrightarrow \mathbb{R}$ be a monotonic function such that $f(I)$ be an interval, then f is necessary continuous on I.*

Proof. WLOG, we can suppose that I is not a trivial interval $I = \emptyset, \quad I = a$ and we suppose that f is strictly increasing. Let $a \in I$, if $a > inf(I)$, we prove that $\lim\limits_{x \to a^-} f(x) = f(a)$ and if $a < sup(I)$, we prove that $\lim\limits_{x \to a^+} f(x) = f(a)$. Indeed let $a \in I$, if $a > inf(I)$, by monotonic limit theorem, there exists a real l such that $\lim\limits_{x \to a^-} f(x) = l$, such that $l < f(a)$. Actually, $l = sup\{f(x), x \in I, x < a\}$, we aim to prove that $l = f(a)$. By contrapositive, suppose that $l < f(a)$, then there exists $l < m < f(a)$, there exists $\alpha \in I$ such that $\alpha < a$. By the hypothesis that f is increasing, we get $f(\alpha) \leq l < m < f(a)$. $f(I)$ is an interval, so there exists $c \in I$ such that $f(c) = m$.

- If $c \geq a$, and f is supposed to be increasing, hence $f(c) \geq f(a) > m$,
- If $c < a$, monotonic limit gives $f(c) < f(a) < m$, which is a contradiction, thus $f(a) = l$.

Similarly, with the right limit. □

Lemma 4.27. *let I be an interval of $\mathbb{R}$ and $f : I \Longrightarrow \mathbb{R}$ be a continuous and injective function. Then f is strictly monotonic.*

Proof. By contrapositive principal, we suppose that f isn't strictly monotone, thus

- $\exists x, y \in I : x < y$ and $f(x) \geq f(y)$
- $\exists x', y' \in I : x' > y'$ and $f(x') \leq f(y')$

The segments $[x, x'], [y, y']$ are defined by $[x, x'] = \{tx + (1-t)x', t \in [0,]\}$ and $[y, y'] = \{ty + (1-t)y', t \in [0, 1]\}$. Then let us define the following functions:

$$\alpha : [0, 1] \longrightarrow \mathbb{R} : t \to tx + (1-t)x', \quad \beta : [0, 1] \longrightarrow \mathbb{R} : t \to ty + (1-t)y'.$$

$\alpha(t) and \beta(t)$ belong to the interval I. Now we consider the function defined by

$$\phi : [0,1] \longrightarrow \mathbb{R} : t \to f(\alpha(t)) - f(\beta(t)).$$

(1) α, β, f are continuous so ϕ,
(2) $\phi(0) = f(x) - f(y) \geq 0$ and $\phi(1) = f(x') - f(y') \leq 0$. The mean theorem value implies that there exists $t_0 \in]0,1[$ such that $\phi(t_0) = 0$, which means that $f(\alpha(t_0) = f(\beta(t_0))$. Contradiction (f is injective). □

The inverse function theorem for strictly monotonic function

Theorem 4.28. *let I be an interval of $\mathbb{R}$ and $f : I \longrightarrow \mathbb{R}$ be a function. Set $J = f(I)$. Then two of the following properties implie the third one.*

1- J is an interval and $f : I \longrightarrow J$ is a bejection function.

2- f is strictly monotonic on I.

3- f is continuous on I.

more; if 1, 2 and 3 are satisfied, then the inverse function $f^{-1} : J \longrightarrow I$ is continuous, strictly monotonic, the same as f.

Proof.
- If 1 and 2 are satisfied then f is continuous (Lemma (4.26)).
- If 1 and 3 are satisfied then f is strictly monotonic (Lemma (4.27)).
- If 2 and 3 are satisfied then J is an interval (MTV theorem). f is strictly monotonic, thus f is injective and by the way bejective.

If 1, 2 and 3 are satisfied the by (Lemma (4.25)), $f^{-1} : J \longrightarrow I$ is strictly monotonic, the same as f. $f^{-1} : J \longrightarrow I$ realize a bejection from J on I, so f^{-1} satisfies 1 and 2, hence f^{-1} is continuous. □

The fact that the domain of f must be an interval is a necessary condition, see for example the following.

Example 4.12. *Let $g : [0,1] \cup]2,3] \longrightarrow [0,2]$ defined by*

$$g(x) = \begin{cases} x, \ \textit{if}\ 0 \leq x \leq 1 \\ x-1, \ \textit{if}\ 2 < x \leq 3 \end{cases}$$

The inverse is

$$g^{-1}(x) = \begin{cases} y, & \text{if } 0 \leq y \leq 1 \\ y+1, & \text{if } 1 < y \leq 2 \end{cases}$$

and it is not continuous because of a jump at $y = 1$.

Theorem 4.29. *Let* f *be a continuous and increasing function on the interval* $[a, b]$. *Let* $\alpha = f(a)$, *and* $\beta = f(b)$. *Then*

(1) The image of $[a, b]$ *by* f *is equal to the interval* $[\alpha, \beta]$ *(*$f([a, b]) = [\alpha, \beta]$*)*

(2) There exists an inverse function $x = g(y)$ *of* f *continuous and increasing on* $[\alpha, \beta]$.

Remark 4.7. *The inverse of a decreasing and continuous function* f *on* $[a, b]$ *is a decreasing and continuous function on* $[\alpha, \beta]$, *where* $\alpha = f(a)$, $\beta = f(b)$. *One can consider the function* $-f$.